Waterslides Operations & Maintenance Manual

VER 2.1

Prepared for:

Project No. 37424 – Poolsider – Group A
Silliman Aquatic Center
Newark, California

January 27, 2018

Reviewed and submitted by: _____
Tim Kwasnicki P. Eng.

www.whitewaterwest.com

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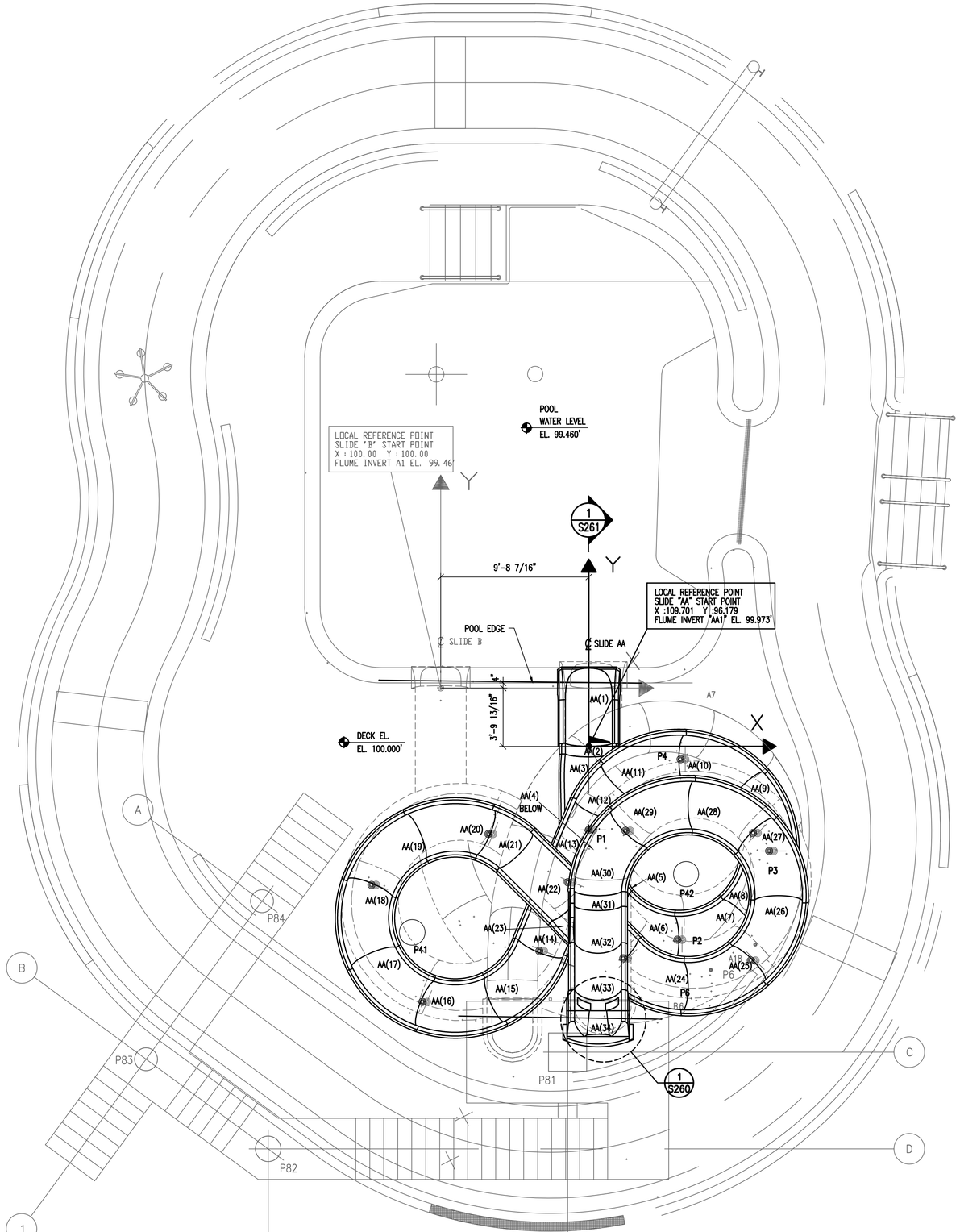

SLIDEPATH LAYOUT

SCALE 1/4"=1'-0"

SLIDE AA CO-ORDINATE TABLE

SECTION NUMBER	PART CODE	DESCRIPTION	ORTN	XCOORD (ft)	YCOORD (ft)	AZIMUTH (deg)	ELEVATION (ft)	GRADE (%)
AA(1)	SMPE-ASSY**	SM Pool Entry with F-Flange		109.701	96.179	270.00	99.973	10.00
	SMPE	Pool Entry						
	SMFALSE	False Flange						
AA(2)	SM15STR-SP[213]	Straight 1500 [213mm]		109.701	95.484	270.00	100.043	10.00
AA(3)	SM15STR-RIGHT	Str 1500 w/ Right End		109.701	90.587	270.00	100.533	10.00
AA(4)	SM9CC-INTRISER	(9000R 2.26') CC Integrated Riser		109.701	89.431	270.00	100.671	14.00
AA(5)	SM4518L	(1800R 45.00') LH	RT	111.431	85.256	315.00	101.321	14.00
AA(6)	SM4518L	(1800R 45.00') LH	RT	115.607	83.526	360.00	101.970	14.00
AA(7)	SM4518L	(1800R 45.00') LH	RT	119.783	85.256	45.00	102.619	14.00
AA(8)	SM4518L	(1800R 45.00') LH	RT	121.512	89.431	90.00	103.269	14.00
AA(9)	SM4518L	(1800R 45.00') LH	RT	119.783	93.607	135.00	103.918	14.00
AA(10)	SM4518L	(1800R 45.00') LH	RT	115.607	95.337	180.00	104.567	14.00
AA(11)	SM4518L	(1800R 45.00') LH	RT	111.431	93.607	225.00	105.217	14.00
AA(12)	SM22518L	(1800R 22.50') LH	RT	110.151	91.691	247.50	105.541	14.00
AA(13)	SM15STR-LEFT	Str 1500 w/ Left End		108.286	87.188	247.50	106.224	14.00
AA(14)	SM15STR-LEFT	Str 1500 w/ Left End		106.421	82.686	247.50	106.906	14.00
AA(15)	SM4518R	(1800R 45.00') RH	RT	103.225	79.490	202.50	107.555	14.00
AA(16)	SM4518R	(1800R 45.00') RH	RT	98.705	79.490	157.50	108.205	14.00
AA(17)	SM4518R	(1800R 45.00') RH	RT	95.509	82.686	112.50	108.854	14.00
AA(18)	SM4518R	(1800R 45.00') RH	RT	95.509	87.206	67.50	109.503	14.00
AA(19)	SM4518R	(1800R 45.00') RH	RT	98.705	90.402	22.50	110.153	14.00
AA(20)	SM4518R	(1800R 45.00') RH	RT	103.225	90.402	337.50	110.802	14.00
AA(21)	SM22518R	(1800R 22.50') RH	RT	105.140	89.122	315.00	111.127	14.00
AA(22)	SM15STR-RIGHT	Str 1500 w/ Right End		108.587	85.675	315.00	111.809	14.00
AA(23)	SM15STR-RIGHT	Str 1500 w/ Right End		112.033	82.229	315.00	112.491	14.00
AA(24)	SM4518L	(1800R 45.00') LH	RT	116.209	80.499	360.00	113.141	14.00
AA(25)	SM4518L	(1800R 45.00') LH	RT	120.385	82.229	45.00	113.790	14.00
AA(26)	SM4518L	(1800R 45.00') LH	RT	122.114	86.405	90.00	114.439	14.00
AA(27)	SM4518L	(1800R 45.00') LH	RT	120.385	90.581	135.00	115.089	14.00
AA(28)	SM4518L	(1800R 45.00') LH	RT	116.209	92.310	180.00	115.738	14.00
AA(29)	SM4518L	(1800R 45.00') LH	RT	112.033	90.581	225.00	116.387	14.00
AA(30)	SM4518L	(1800R 45.00') LH	RT	110.303	86.405	270.00	117.037	14.00
AA(31)	SM15STR-SP[300]	Straight 1500 [300mm]		110.303	85.430	270.00	117.173	14.00
	SMR15STR-SP[300]	Straight 1500 Riser [300mm]						
AA(32)	SM4CC	(4000R 12.62') CC		110.303	82.634	270.00	117.885	37.57
	SME4CCL	CC End LH						
AA(33)	SM15STR-SP[1000]	Straight 1500 [1000mm]		110.303	79.563	270.00	119.039	37.57
AA(34)	SMENTRY-ASSY**	SM Entry with plates		110.303	76.505	270.00	119.082	0.00
	SMENTRY	Entry Section						
	SMPLATESET	Injector Cover Plate Set						

Total ride length: 140.103 (ft)

Total number of parts: 38

Total SM ride length: 140.103 (ft)

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BOM Notes: Moved slides about 4" closer to slide AB to avoid column and match surveyed distance.

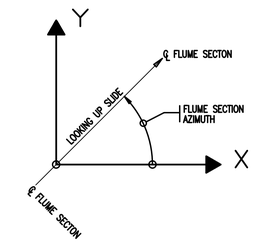
Slide Information details

Silliman Aquatic Center, Newark, CA, Project # 37424

	1
Ride Serial Number	37424-AA
Date of Manufacture	2018
Ride Name	Poolsider
Product Family	SM
Ride Speed	maximum 22.0 feet per second
Water Flow	design 800 US gallons per minute
Passenger Capacity by Weight	300 pounds
Passenger Capacity by Number	one person
Minimum Rider Height	48 inches
Vehicle Type	no vehicles
Patent Number	www.whitewaterwest.com

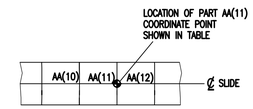
NOTES:

1. FOR GENERAL NOTES AND SPECIFICATIONS SEE DWG 130.



FLUME SECTION DIAGRAM

NOT TO SCALE



PART COORDINATE LOCATION

NOT TO SCALE

REV	REVISION DESCRIPTION/ISSUE RECORD	DATE	PE
0	ISSUED FOR CONSTRUCTION	2018-01-23	
1	ISSUED FOR INFORMATION	2018-01-22	

DESIGNED BY IR	DATE 2018-01-11	PROJ. ENG. CHECK	DATE
DRAWN BY MT	DATE 2018-01-11	DRAFTING CHECK	DATE
SIZE D	SCALE PER VIEW	ARCHIVE FILE No.	37424A200

NOTE:- IMPERIAL DIMENSIONS
ALL ELEVATIONS AND CO-ORDINATES ARE IN FEET
ALL OTHER DIMENSIONS ARE IN FEET AND INCHES

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GROUP A
SLIDEPATH LAYOUT
AND COORDINATE TABLE

SILLIMAN AQUATIC CENTER
NEWARK, CA, UNITED STATES

PROJECT No.	DRAWING No.	REVISION
37424	A200	0

PART I
GENERAL OPERATING GUIDELINES

Part I GENERAL OPERATING GUIDELINES

Important Notice
The following operating procedures are presented to waterslide owners as guidelines only. They are not intended to be mandatory or exhaustive. These guidelines do not replace proper consultation with designers, manufacturers and legal counsel, and compliance with local, state (provincial) and federal laws and regulations involving, but not limited to: health, safety and building codes.

General Operating Considerations

1. Customers

While customers are the key ingredients to the success of any waterpark, certain rules and procedures must be made clear for the park to function efficiently and safely.

1.1 Posted Signs

A sign must be posted at the entrance to the waterpark that states user responsibility and conveys the following rules and regulations:

1. Read and obey all posted signs.
2. Obey all instructions given by the attendants and operators.
3. No glass, cigarettes and/or food are permitted in the water slide area.
4. Anyone under the influence of alcohol or drugs will not be permitted in the park.
5. Pregnant women and those with a history of heart problems or back trouble should not use the slides.
6. Failure to follow rules and instructions can lead to serious injuries.
7. Ride at your own risk.

Note: Specific sign requirements for each waterslide are detailed in Part II - Specific Operating Guidelines.

1.2 Access

Provisions must be made to prevent customers from entering the water slide at any points other than at the designated entry access points.

1.3 Physical Condition

Waterslide customers must be checked for proper swimwear and adequate physical condition. Swim attire should not have any metal parts that could damage the slides or cause injuries. Customers who show signs of drug and/or alcohol impairment will not be permitted on the site.

1.4 Visitors and Spectators

Visitors and spectators should be physically separated from the areas used by sliders. They should be kept a safe distance away from any wet areas and not be permitted to interfere with the sliders.

1.5 Food and Drinks

No food or drinks are allowed in the immediate area of flumes and pools or on decks, stairs and walkways surrounding flumes and pools. Food and beverages should be confined to visitor and spectator areas or similarly marked areas for sliders. Trash containers should be provided to keep litter off decks and walkways and out of flumes and pools.

2. Attendants

Waterpark attendants are responsible for safe and orderly conduct of customers, safe entry and exit of riders from the slides, general crowd control and emergency procedures.

During operating hours, there must be at least one attendant on duty at all times who has completed a Red Cross or equivalent Standard First Aid and Safety course. This person must also be competent in carrying out any emergency procedures specific to the slide he or she is operating.

Every water slide should be equipped with a battery or electronically operated system to provide voice communication between attendants at different locations on the slides and park office or permanently manned control center.

2.1 Top of Slides

The area at the top of every water slide must be supervised by at least one attendant who has continuous and direct supervision of that area and is responsible for:

- Ensuring that customers conduct themselves in a safe and orderly manner.
- Ensuring that customers enter slides or rafts/mats properly.
- Ensuring that customers are dispatched at safe intervals.
- Visually surveying all areas of the water slide that are visible from attendant's position, with the exception of the splash pool, and communicating any problems to the other attendants.
- Ensuring that customers meet the minimum and/or maximum restrictions for the slide.
- Ensuring that any vehicles used are in good condition and if inflated, inflated properly.

2.2 Splash Pool

The recommended depth for the splash pool is 3.5 feet (1.07 metres) except otherwise noted in special conditions and therefore must be supervised by at least

one certified lifeguard who has continuous and direct supervision of that area and is responsible for:

- Ensuring that customers conduct themselves in a safe and orderly manner.
- Ensuring that customers move into and out of splash pool in a quick and orderly manner.
- Controlling running, horseplay or other unsafe behavior on pool decks.
- Ensuring pool water levels are correct and as per marked water levels. Correct operating water level is critical to safe operation.

2.3 Runout Lane

The runout lanes of waterslides must be supervised by at least one attendant who has continuous and direct supervision of that area and is responsible for:

- Ensuring that customers conduct themselves in a safe and orderly manner.
- Ensuring that customers exit runout lane in a quick and orderly manner.
- Controlling running, horseplay or other unsafe behavior on slide apron.
- Ensuring water levels are at level marks and flow transition is within indicator markings. Correct operating water levels are critical to safe operation.

2.4 Water Treatment

A specific person on each shift should be responsible for chlorination and water treatment procedures. They should be thoroughly trained in all aspects including, routine operations, handling, connection, disconnection, emergency procedures and leak control procedures. A safety chart should be posted in or near the chlorination room and a second safety chart should be placed in the pool office near the telephone. Emergency telephone numbers should also be shown on the safety charts.

2.5 Pool Operations

A specific person on each shift should be made responsible for all swimming pool operations, such as: circulation and filtering system operation, maintenance, inspections, backwashing and cleaning. They should be thoroughly trained by an expert in swimming pool operations.

3. Inspections

Permanently bound log books should be kept to record all inspections, operational tests, water quality monitoring, all accidents and unusual occurrences. Log books should be kept for a period of at least six (6) years from the date of the last entry. Log book entries that cover the twelve (12) month period previous to the last entry in the log book should be kept readily available.

3.1 Slide Inspections

Daily inspections must be made to check for:

- obstructions in slide paths
- cracks, chips or bubbles in sliding surfaces
- rough patchwork at joints or cracks
- caulking protruding from joined flanges
- leaking seals at joints
- loose risers on turns
- excessive movement of flumes when walked on
- joints opening up
- proper inflation and condition of rafts, tubes and vehicles
- landing or pool bottom padding, if used, is in good condition and properly secured.
- There are no dirt or chemical buildups such as calcium on the sliding surfaces that may affect the slide's performance.

3.2 Mechanical Inspections

Daily inspections must be completed on each waterslide:

- before water is circulated to ensure no safety hazards exist
- with the water circulating to ensure there is sufficient water flowing in the channel and correct water level is in splash pool or runout lane
- Flow meters are operating and valves have not been tampered with.
- Water levels in pool or shutdown lanes are correct and as per markings.
- Water flows are as per manufacturer's specifications.
- to ensure that all the pumping and filtration equipment is operating correctly
- to ensure that conveyor system is operating properly and poses no hazard

It is the attendant's responsibility to ensure water flow is maintained in accordance with manufacturers' guidelines. Serious hazards may develop when flow is interrupted in a flume.

3.3 Chemical Inspections

Proper water chemical balance must be obtained each day before the facility is opened to the public. While the facility is operational, water quality testing should be performed at intervals in accordance with local health authority's guidelines. Inspections should also be made during periods of heavy usage to ensure circulation and filtration systems are handling peak loads. The attendant must keep logs of all water testings performed. A typical form for water quality testing is shown in Appendix 1-I.

The attendant must ensure that customers cannot come in contact with water treatment, re-circulation and filtration equipment. Mechanical and chemical storage rooms must be locked at all times to prevent customer access.

4. Accidents

The most common accidents in waterparks are: slips and falls, collisions between one riding customer and another customer, impact with the pool surface, and abrasions from sidewalls. Operators should keep detailed logs of all reported accidents onsite. A typical accident report form is included in Appendix 1-II.

4.1 Incident Reporting

The owner or operator shall notify the manufacturer (Whitewater West Industries Ltd.) of any known serious incidents that require immediate hospital admission. Minor injuries that may or may not require treatment shall be reported to the manufacturer if deemed to be related to the equipment design or function.
There may be local jurisdiction and laws requiring notification of incidents. The owner or operator may be required to notify these local jurisdictions in addition to the above.

4.2 Floatation Devices

The operator should not infer that rafts, tubes or any other floatation device, with exception of government approved life jackets, offer the user any safety benefit. Where unapproved floatation devices are provided, signs should be posted stating they are for pleasure only and should not be used as a substitute for approved floatation devices.

4.3 Slips and Falls

Slips and falls are a constant concern in all areas of the waterpark. All surfaces intended for pedestrian traffic, whether in the water, on walkways, on loading platforms or on stairs should be made and maintained slip-resistant. Proper periodic maintenance of walking surfaces should be carried out to remove the growth of algae and fungi which could inhibit the surfaces intended function. Adequate drainage should be maintained around all areas of the slides.
The entry section to flumes should be given particular attention with respect to slips and falls. The combination of the flowing water and the customer contacting water on a slippery surface for the first time can be confusing and disorienting. The exit area of the splashdown pool will always be wet and flowing with water which requires special attention with respect to slip-resistant surfaces, drainage and number and location of attendants.
To minimize slip and fall type accidents, attendants throughout the waterpark should control running and horseplay.

4.4 Bodily Collisions

Bodily collisions occur in flumes and splash pools as a result of riders traveling at different speeds and riders slowing or stopping in the flume. Restricting traffic flow with proper dispatching intervals can minimize these bodily collisions. The top attendant and lower attendant must have a view of much of the slide and a means of

communication between them. The entry attendant must be alert and aware that people will travel at different rates due to variations in body weight, body friction, and rider position. Generally, the heavier the person, the faster the rider will travel. Slippery materials like nylon slide much faster than cotton clothing. All other factors being equal, riders in a sitting position will travel slower than a person lying flat on their back. Under no circumstances should customers be permitted to ride flumes in a standing position. Similarly, running or leaping entry into flumes is not to be permitted.

In addition to slide entry attendants, use of signal lights or electro-mechanical starting gates should be considered for rider entry control.
As bodily collisions in splash pool area cannot be fully controlled even with proper entry rates; splash pool attendants should be well trained, attentive and aggressive, continuously urging and encouraging customers to leave the flume discharge and splash pool area as quickly as possible. They must be alert to the possibility of problems and be prepared to enter the water at any time to assist a customer. In the event of a delay at discharge, following riders should be stopped at the entry. Riders tend to congregate in the pool because they are temporarily confused and disoriented upon impact with the water, to wait for their friends who are coming behind, or to socialize with persons in the splash pool. Collisions also result from multiple rider entry (i.e. trains). This practice is absolutely prohibited even for small children being carried on their parent's legs.
If marker buoys or lane ropes are used in the splash pool, they must be arranged so they do not become obstacles for riders. Such devices, if used, should float and be simple in design to avoid the possibility of entangling riders.

4.5 Chemical Hazards

Personnel must be fully trained in the use of water chemicals. Potentially fatal accidents could result from the release of gaseous chlorine. Improper chemical control could result in skin and eye irritation from excessive treatment. Bacterial problems could also arise from inadequate treatment.

5. Emergency Procedures

Every park must have an emergency plan with specific procedures covering:

- crowd control
- park evacuation
- drowning
- heart attacks
- cuts and burns
- neck and spinal injuries
- power or other utility failures
- fires
- security (fights, robbery, vandalism)

- exposure to chlorine
- environmental conditions (e.g., lightning, windstorms, hail, earthquakes)

An emergency plan should address those factors which affect, human safety, protection and correction of equipment, and customer relations. All employees should be trained and drilled periodically in the execution of the emergency plan. The emergency plan should be easily accessible and be located with first aid equipment and emergency telephone numbers.

Each facility should have a list of current emergency numbers posted by a telephone. The list should include the contact information for the nearest available doctor, ambulance service, hospital, rescue service, police department, fire department and poison control center. A typical emergency telephone list is shown in Appendix 1-III.

Each facility should have available the following first aid supplies:

- A standard 24-unit first aid kit stocked and readily accessible for use.
- A stretcher and blankets.
- A standard plywood backboard or other acceptable splint for persons with back and neck injuries.
- An area or room set aside for the emergency care of injuries.

6. Operator's Changes

WhiteWater should be advised of any changes made in slides, structures or equipment, manufactured, designed or installed by WhiteWater West Industries Ltd. Changes could affect the safety of the attraction.

PART II
SPECIFIC OPERATING GUIDELINES

Part II SPECIFIC OPERATING GUIDELINES

POOL SIDER (SM) into Pool – Slide AA

The primary operating concern in Pool Sider Slides is the possibility of one riding customer colliding with another customer, either in the flume or in the splashdown pool. This type of occurrence can be minimized by the use of well-trained attendants controlling entry and exit conditions. Other safety considerations include: slips and falls in the entry and exit areas, abrasion from the sidewalls, and impact with the splashdown pool surface.

1. Posted Signs

1.1 Entry Station

Signs should be posted at the entry station of the ride to convey the following rules and regulations:

1. Maximum operational load: 1 person, 300 lbs (136.4 kg).
2. Children under 48 inches (1.22 m) tall are not permitted on this ride.
3. Eyeglasses must be securely affixed to riders with head straps.
4. Swim wear with exposed zippers, buckles, rivets or metal ornamentation are not permitted.
5. Only one rider to enter the flume at a time. Single Riders only!
6. Never form chains.
7. Slide must be ridden feet first lying on your back or in a sitting position (sit up to go slower, lie down to go faster).
8. Riders must wait for the attendants start signal before starting the ride.
9. Keep arms and hands inside the flumes at all times.
10. Do not run, dive, stand, kneel, rotate or stop in the slide.
11. At the end of the slide, obey all instructions by splash pool attendant and exit quickly.
12. CAUTION: For safety reasons, pregnant women and persons with heart conditions or back trouble should not ride this slide.

1.2 Splash Pool

Signs should be posted at the splash pool of the ride to convey the following rules and regulations:

1. Do not block the end of the slide.
2. Leave the splash pool quickly and orderly.

3. No swimming or flotation devices allowed in the splash pool.

2. Attendants

Attendants should be alert to controlling crowd behavior, ensuring safe and orderly entry and exit and controlling the proper entry rate into the slide.

2.1 Attendant / top (start of ride) - 1 attendant per slide

This requirement may be relaxed to not less than one attendant per platform area or level, provided that the operator ensures that attendants are able to fulfill all duties as listed.

Duties:

1. Ask each rider if they know how to ride the ride.
2. Assure the rider is in the correct body position when ready.
3. Control dispatch of riders one at a time and shall not dispatch a rider until the flume is clear of obstruction and any previous rider.
4. Ensure all riders meet requirements.
5. Ensure eyeglasses are securely affixed to the riders.
6. Observe bathing suits for zippers, buckles, rivets, etc. and do not allow them on the slide.
7. Observe riders for any physical impairments that compromise safety and do not allow them on the slide.
8. Ensure the crowd behaves in a safe and orderly manner.
9. Monitor riders in areas visible from assigned positions.

2.2 Attendant / splash pool: 1 lifeguard per 4 slides

Duties:

1. Ensure customers conduct themselves in a safe and orderly fashion.
2. Do not allow riders to block the end of a slide. Delay dispatch of next rider if the slide landing area is obstructed.
3. Ensure that all riders exit the pool immediately upon splashdown. Do not permit guests to sit on pool steps or congregate around pool edge.
4. Be prepared to enter splash pool to assist customer out of pool.
5. Do not allow any swimming or flotation devices in splash pool.

NOTE: Special attention is required at the splash pool due to reduced visibility from flowing water and the splash caused by riders entering the pool. The clarity of quiescent splash pool water is normally acceptable if no turbidity is noted and the bottom detail is clearly visible.

3. Vehicles

No vehicles are used in this slide. Patrons ride the slide in bathing suit attire according to instructions above.

4. Water Flow

The water flow should be maintained at approximately 800 U.S. gallons per minute (50.5 litres/second) unless directed by Whitewater representative.

PART III
MAINTENANCE GUIDELINES

Part III MAINTENANCE GUIDELINES FOR FRP FLUME

If properly maintained, your waterslide will give you years of service and keep sliding the way it was meant to - SAFE and FUN. By keeping the slide and its components properly maintained, you will be ensuring high performance and long life.

1. Cleaning

Maintenance requires removal of surface dirt, grease and suntan oil by cleaning the flumes with a soft-bristled brush and mild detergent such as 3M - “Multi Purpose Boat Soap”. For hard water buildup (a white chalky discoloration) removal/maintenance use “C.L.R.” or other cleaners specific for this purpose. For removal of difficult stains use a cut polish or rubbing compound and finish wax or a combination product such as 3M “Marine One-Step Fiberglass Restorer and Wax”.

2. Waxing

At least once for every two months of operation, the fiberglass finish should be waxed and buffed with high quality paste wax or polish such as 3M “Marine One-Step Fiberglass Restorer and Wax”. On scratched, dull or faded areas use a fine buffing compound to restore the original gloss before waxing. Prolonged exposure to the ultra-violet rays of the sun can in time, cause discoloring and fading. We recommend using paste waxes or polishes containing ultra-violet screens. The more often the surface is waxed, the longer it will last.

Washed and waxed surfaces can be critical to proper and safe operation. Calcium buildup, for example, may cause vehicles to stick, which may lead to unpredictable behavior and lead to an unsafe condition.

CAUTION: If you use power tools for buffing or polishing, use extreme care to prevent overheating the surface. Overheating a rubbing compound burns small gritty particles into the slide surface causing blistering and streaking. This may require the area to be sanded and repainted.

3. Joints

Slide joints play an important role in rider comfort and pleasure. Proper maintenance will keep the joints smooth and trouble free for years.

3.1 Joint Caulking

All fiberglass sections will expand and contract with temperature changes. Therefore, joints between sections are filled with a flexible, elastic caulking. Should the joints need re-caulking, all the old filler should be removed, the fiberglass should

be cleaned and joints should be re-filled using a flexible polyurethane sealant. **DO NOT** use plastic filler that will harden. **Under no circumstances should the joints be fiberglassed over.** Excess joint caulking should be removed from the slide surface by scraping it off with a putty knife. Any remaining traces should be wiped off with lacquer thinner applied to a rag. Do not pour lacquer thinner directly onto a flume joint as it can dissolve the sealant.

3.2 Joint Roughness

Joints may become rough with use, especially if metal articles are allowed down the slide. For minor chips and cracks, the joints may be fixed using the procedures outlined in Part III, Section 4.
More serious cases may require power sanding that will cut through the gelcoat. This will require a spray application of new gelcoat by a trained person, finished by sanding and polishing and waxing as outlined in Part III.

4. Patching

Chips and cracks may appear in fiberglass flumes from normal usage. The following sections outline procedures to repair minor damage to slides. Due to the hazardous nature of materials used and variability of application methods, we recommend that a qualified fiberglass laminator perform the repairs.

4.1 Minor Chips and Cracks

In areas that have chips or cracks in the gelcoat only, or have a scratch that is deep enough to penetrate through the gelcoat to the fiberglass but not deep enough to go completely through the laminate, follow the procedures given in Part III, Section 4.4.
NOTE: For damage that penetrates completely through or deep into the laminate, we recommend that a qualified fiberglass laminator perform the repairs.

4.2 Surface Cracks

Hairline cracks, sometimes called spider webbing, or star cracks may develop in the gelcoat or surface coating of the fiberglass product. This is caused by a combination of weathering, vibrations, and/or impacts. Although unsightly, they do not necessarily affect structural strength. The hairline cracks can be fixed by sanding out the affected area with 100 grit sandpaper and re-coating the surface with gelcoat. Follow the patching instructions in Part III, Section 4.4.3, omitting steps 4 through 6.

4.3 Cracks around Flanges

The area around the flanges of flumes contains a thick layer of gelcoat which is very stiff. Stresses from normal use may cause cracks in the gelcoat around the flanges. These are not structural cracks. Flange cracks may be remedied by sanding the affected area with 100 grit sandpaper and re-coating with gelcoat. Follow the patching instructions in Part III, Section 4.4.3, omitting steps 4 through 6.

4.4 Repair Procedures

WARNING:

Patching kits contain toxic, combustible and corrosive elements. Use only under well ventilated conditions. Do not inhale fumes. Seek medical attention if polyester resins are swallowed or inhaled. Should contact occur with the eye flush with water for 15 minutes and seek medical attention. If person is wearing contact lenses and the lenses did not flush out from the water, have the person try to remove AFTER the flushing procedure. If contents come in contact with skin or clothing, clean the area immediately with acetone.

Before you start, you will need:

1. Sandpaper: 36 - 100 grit; 220 - 320 grit wet and dry; 600 grit wet and dry
2. Sanding block
3. Acetone
4. 2" masking tape
5. Utility knife
6. Putty knife
7. Mixing board
8. Mixing cup
9. Fiberglass resin hardener (catalyst), use a peroxide based catalyst (MEKP)
10. Patching compound
11. Gelcoat paint
12. 2" fine paint brush
13. Automotive cut polish
14. High quality paste wax such as "Turtle" wax

4.4.1 Storage Instructions

Store uncatalyzed patching compound and gelcoat in closed metal containers in a cool area below 73° Fahrenheit (23° Celsius) out of direct sunlight. Do not allow to freeze.

Store liquid catalysts in their original containers according to manufacturer's instructions.

4.4.2 Mixing Instructions

Mix only small amounts at one time. The mixtures will harden in twenty (20) minutes to 1 hour, depending on the temperature and the amount of catalyst added. The mixture will harden faster at higher temperatures and with increasing amounts of catalyst added. Use caution when adding catalyst, over catalyzing will result in the mixture cracking. Do not pour catalyzed mixtures back into the can. Do not reduce materials with a "conventional" paint or lacquer thinner.

Patching Compound: Add one (1) part catalyst to one hundred (100) parts patching compound on the mixing board. Mix well.
Gelcoat: Add one (1) part catalyst to one hundred (100) parts gelcoat in the mixing cup. Mix well.

4.4.3 Patching Instructions

1. Sand area to be patched with 36 -100 grit sandpaper. Feather sand to moulded surface and remove all rough edges.
2. Wash area clean with acetone and allow to dry.
3. Mask off area to be patched with 2" masking tape.
4. Using a putty knife, spread the patching compound smoothly over the prepared area to the top of the tape.

NOTE: Area must be dry and clear of contaminants, such as dirt and sanded particles, etc.

5. Keep the patched area dry and allow 20 minutes to 1 hour to harden.
6. After the patch has hardened, sand the patched area flush to the surrounding surface with 100 grit sandpaper.

NOTE: Sanding will remove the masking tape. Re-tape the area before painting.

7. Using a 2" fine paint brush, paint the area with gelcoat. Allow twenty (20) minutes to one (1) hour to dry.

NOTE: The paintbrush and tools can be cleaned with acetone.

8. After the gelcoat has dried, remove the masking tape. Using a sanding block, wet sand with 220 - 320 grit sandpaper. Repeat with 600 grit sandpaper.
9. Buff painted surface with auto cut polish and wash clean with water.
10. Wax and buff painted surface with a high quality paste wax.

PART IV
SUPPLEMENT TO MANUAL FOR ASTM STANDARDS

PART IV SUPPLEMENT TO MANUAL FOR ASTM STANDARDS

1. The ASTM standard F 2376-XX Standard Practice for Classification, Design, Manufacture, Construction, and Operation of Water Slide Systems.

The ASTM standard F 2376 for Water Slide Systems is a standard developed specifically to cover Water Slide systems. The standard references several other standards which together form the basis of a complete standard for the Design, Manufacture, Construction and Operation of Water Slide Systems. Operators are encouraged to become active members of the ASTM F24 group. The fees are nominal and provide access to the relevant standards. Operators should be aware of these standards and the responsibilities outlined within them.

General Description of Rides
Waterslides use the flow of water to propel or lubricate the movement of riders down a fixed fiberglass flume section. The riders may ride alone with no vehicle, in groups on specially designed inflatable inner tubes or on a custom made inflatable boat or raft. The riding configuration depends on the specific product and the method of riding.

Description of Motion
Waterslide flumes are fixed in position and do not move. Propulsion is from gravity aided by water flow in the flume. Riders with or without vehicles will experience motion and accelerations consistent with the flume path. Movements include right and left turns, vertical curves, and deceleration at the finish. Riders or vehicles shall maintain constant contact with the flume ride surface.

Operational Restrictions Due to Weather
Wind - Waterslide structures are designed to withstand wind loads specified in the building code in force for the ride locations. WhiteWater recommends suspending operation during sustained winds in excess of 40 mph (64 kph).
Lightning - Virtually all waterslides use elevated towers as starting platforms. Operation of rides should be suspended during any storm or weather conditions where lightning may occur.
Rain - Rain will have no effect on the operation of the slides, with one exception. All flumes rely on the cushioning and braking action of the specified water flow. Rain water can create a very slippery surface and without the braking action of the larger water flow can create an extremely hazardous condition. Under no circumstances should the slides be operated without the main water supply running at specified capacity.

Safe water flow
The water flow rate is important to the safe operation of the ride. Operators shall make sure that the water supply is adequate. Insufficient water volume can result in higher speed. The water tends to restrain and cushion motions and impacts as well as reduce speed.

Evacuation and Emergency Procedures
In an emergency situation affecting the slide:
1. Top attendants shall cease dispatch of all riders.
2. If water flow is impeding the remedy of the emergency situation, use E-stop to turn off power to the pump(s).
3. Riders in the slide shall continue down the slide and exit normally if possible.
4. Attendants shall ensure all riders have exited the slide either through the bottom or at egress locations.
5. Patrons may remain queued on tower and stairs if they are not affecting the emergency situation.
6. If necessary to evacuate the tower, patrons shall be guided by top attendant to exit via the stairs from the start tower or egress location.
7. Slide should remain closed until cause of emergency situation is investigated and resolved.
8. In case of serious injuries or problems, operator should notify manufacturer and other regulatory bodies as required by law.

2. The ASTM standard F 770 - XX Standard Practice for Maintenance Procedures for Amusement Rides and Devices.

The ASTM Standard F 770 for Amusement Rides and Devices is a broad-based standard covering all the various rides and attractions encountered in the amusement industry. Most of the requirements are directed at complex mechanical devices that are subject to wear. These devices may have critical connectors and linkages that must be inspected regularly to avoid catastrophic failure. Waterslides in general are quite simple. The following is WhiteWater's requirement for inspection of waterslides to conform to ASTM. These are to be used in conjunction with WhiteWater's Operation & Maintenance Manual.

Operational Tests of Critical Operating Items
Waterslides use water pumped into the starting area of the ride to operate. The water volume is important to safe operation of the ride. In general, the less water volume pumped, the faster the rider will travel. If during operation, the water supply is restricted or fails, operators shall suspend operations when water volume is below specified operating requirements. A basic operational test should be performed daily in accordance with the Waterslides Operations & Maintenance Manual.

Inspection of the fiberglass and sliding surface must be made daily in accordance with the Waterslides Operations & Maintenance Manual. Small chips or surface imperfections may cause injury to riders.

Recommended Maintenance/Inspection

The daily inspection of the slides is detailed in the Waterslides Operations & Maintenance Manual Part I - Section 3 and Part III - Section 1, which covers the operational components of the slide. The only other items that should be inspected are the supporting structures for the slides and start platforms.

Inspection of supporting structures:

- Steel structures that are hot dipped galvanized will require visual inspections on a semiannual basis. Areas around field welds should be carefully checked and touched up with zinc rich paint as necessary. If rust streaking occurs, the source should be identified and attended to. If steel structures are ten years old or greater, a comprehensive professional inspection should be performed.
- Steel structures that are painted without galvanizing should be visually inspected on an annual basis. If steel structures are five years old or greater, a comprehensive professional inspection should be performed.
- Wood structures should be visually inspected annually for integrity, tightness of fasteners and general condition. If wood structures are over five years old, a comprehensive professional inspection should be performed.

Climactic and operation conditions vary greatly and will affect the interval times above. In more corrosive environments such as seaside or some indoor installations, the interval times should be reduced.

Other routine checks completed annually:

- All concrete filled stairs and decks should be visually inspected for worn finishes on the nosing and cracked concrete. The underside of the metal decking should be checked for deterioration.
- All welds should be visually inspected for condition and all bolts should be snug.
- Anchor bolts should be checked for tightness and corrosion. Grout around base plates should be in good condition.
- Inspection and maintenance of fiberglass should be performed as per the WhiteWater Operations & Maintenance Manual Part III.

Comprehensive professional inspections should be performed by a qualified engineering or inspection firm. They perform inspections with methods they deem appropriate to assure accurate review of the structure. These methods may include non-destructive testing using ultrasound or other techniques. Inspection would include condition of metal components, welds, bolts, and any other critical items. The inspecting agency should produce a certificate of inspection and/or a report on condition for review by a qualified engineer.

Fiberglass Wear Limits
The fiberglass should be inspected regularly as per the WhiteWater Operations & Maintenance Manual Part I - Section 3 and Part III - Section 1. The riding surface of the flume is subject to wear and weathering. The exterior of the slide is subject to weathering. The fiberglass should be inspected professionally if five years old or older.

Other Wear Limits
Rides using boats will experience wear on the boats. If the wear is excessive it may impair the function of the boat and operation of the ride. This is especially true if the ride uses shutdown lanes. The wear should be checked in accordance with the boat manufacturer's recommendations.

Replacement Fasteners
All replacement fasteners shall conform to the standards specified in the engineering drawings.

Electrical / Mechanical Systems Operation and Maintenance
All electrical motors, pumps, controls etc. shall be inspected and maintained in accordance with their manufacturer's recommendations.
All pump intake grating or screens must be inspected to ensure that no structural flaw is evident.
Note: Whenever inspection or maintenance work is being done on electrical/mechanical systems, including pump intake grating, all related equipment must be electrically isolated using recognized equipment lock out procedures.

Appendix 1-I

TYPICAL DAILY WATER QUALITY LOG

Location: _____ **Date:** _____

Weather: _____

WATER TEST				FILTER CHECK		
Time	***Chlorine Count***	***pH***	***Water Clarity***	***Backwash***	***Temperature***	***Initial***
10:						
11:						
12:						
13:						
14:						
15:						
16:						
17:						
18:						
19:						
20:						
21:						
22:						
23:						
24:						

Appendix 1-II

TYPICAL FIRST AID REPORT

First Name of Victim:	**Middle Initial:**	**Name of Person Filling Report:**	
Last Name:		**Location Address:**	
Mailing Address:		**City:**	**State/Province:**
City:	**State/Province:**	**Zip/Postal Code:**	**Location Phone #:**
Zip/Postal Code:	**Phone #:**	**Location Name:**	

Date (MM/DD/YY) and Time of injury	(month)	(day)	20	, at	a.m. / p.m.
Date and Time Injury Reported to First Aid Attendant	(month)	(day)	20	, at	a.m. / p.m.
Location of Accident:					
What Happened – Victim Statement:					

Name of Witness:		
Witness Mailing Address:		
City:	**State/Province:**	**Postal Code:**
What Happened – Witness Statement:		

First Aid Attendant Description of Injury:		
Was an Ambulance called (circle) Yes / No	**Manager on Duty:**	
Manager Statement:		

I hereby certify that the above statement is true to the best of my knowledge.

Victim Signature	**Witness Signature**	**First Aid Attendant Signature**	**Manager on Duty Signature**

Appendix 1-III

EMERGENCY PHONE NUMBERS

Location Name:		**Location Phone:**	
Location Address:			

DIAL 911 for Police, or Medical Emergency (if available in your area)

EMERGENCY SERVICES			
Police:		**Sheriff:**	
Hwy Patrol:		**Ambulance:**	
PLUMBER		**VENDING MACHINE COMPANY**	
Company Name:		**Company Name:**	
Contact Name and Phone:		**Contact Name and Phone:**	
Secondary Contact Name and Phone:		**Secondary Contact Name and Phone:**	
GENERAL CONTRACTOR		**WATERSLIDE MANUFACTURER**	
Company Name:		**Company Name:**	**WhiteWater West Industries Ltd.**
Contact Name and Phone:		**Contact Name and Phone:**	604-273-1068
Secondary Contact Name and Phone:		**Secondary Contact Name and Phone:**	
ELECTRICIAN			
Company Name:		**Company Name:**	
Contact Name and Phone:		**Contact Name and Phone:**	
Secondary Contact Name and Phone:		**Secondary Contact Name and Phone:**	
SWIMMING POOL SUPPLY			
Company Name:		**Company Name:**	
Contact Name and Phone:		**Contact Name and Phone:**	
Secondary Contact Name and Phone:		**Secondary Contact Name and Phone:**	

WHITEWATER

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